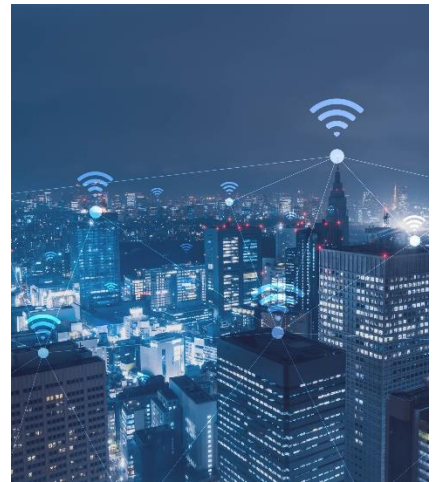
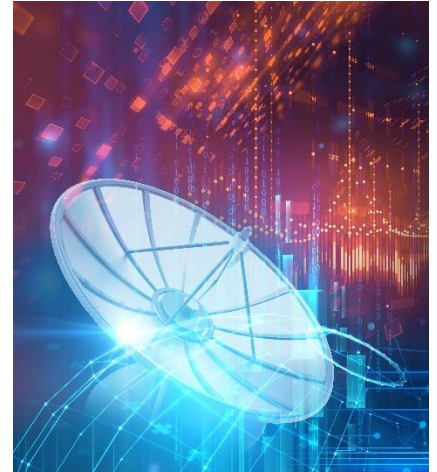


# Model 825-M Ultra-Agile Signal Source



## Features

- Phase Coherent
- Fast Switching as Low as 5  $\mu$ s
- FM, Chirps, Pulse
- Internal OCXO, External Variable Reference

## Applications

- ATE
- LO for Frequency Converters
- Telecom / Satellite Com



## Model 825-M

8 kHz to 20 GHz Ultra-Agile Signal Source

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## DEFINITIONS

- The specifications in the following pages describe the warranted performance of the instrument for  $23 \pm 5$  °C after a 30-minute warm-up period (unless otherwise stated).

**Min/Max:** Parameter range that is guaranteed by product design, and/or production tested. Warranted performance specifications include guard-bands to account for the expected statistical performance distribution, measurement uncertainties, and changes in performance due to environmental conditions.

**Typical:** Expected mean values, not warranted performance.

## INTRODUCTION

- The Model 825-M-20 is a very compact, very agile signal source series with frequency range of up to 20 GHz. It combines fast switching speed with low phase noise and good signal purity. The single-channel unit is available as flange- and rack-mountable module or in a compact desktop enclosure with front panel control.

The multi-channel version Model 825-M-20-X is available in 1, 2, 3 or 4 channel configurations in a standard 1U 19" rack-mountable enclosure. For high phase coherence, RF channels are locked to a common frequency reference. The Model 825-M-20 has standard communication ports USB and ETHERNET, and optionally GPIB. All communication ports support the SCPI 1999 command set. The Model 825-M-20 also features an FCP (Fast Control Port) allowing for ultra-fast user-controlled list sweeping and frequency hopping.

## SPECIFICATIONS

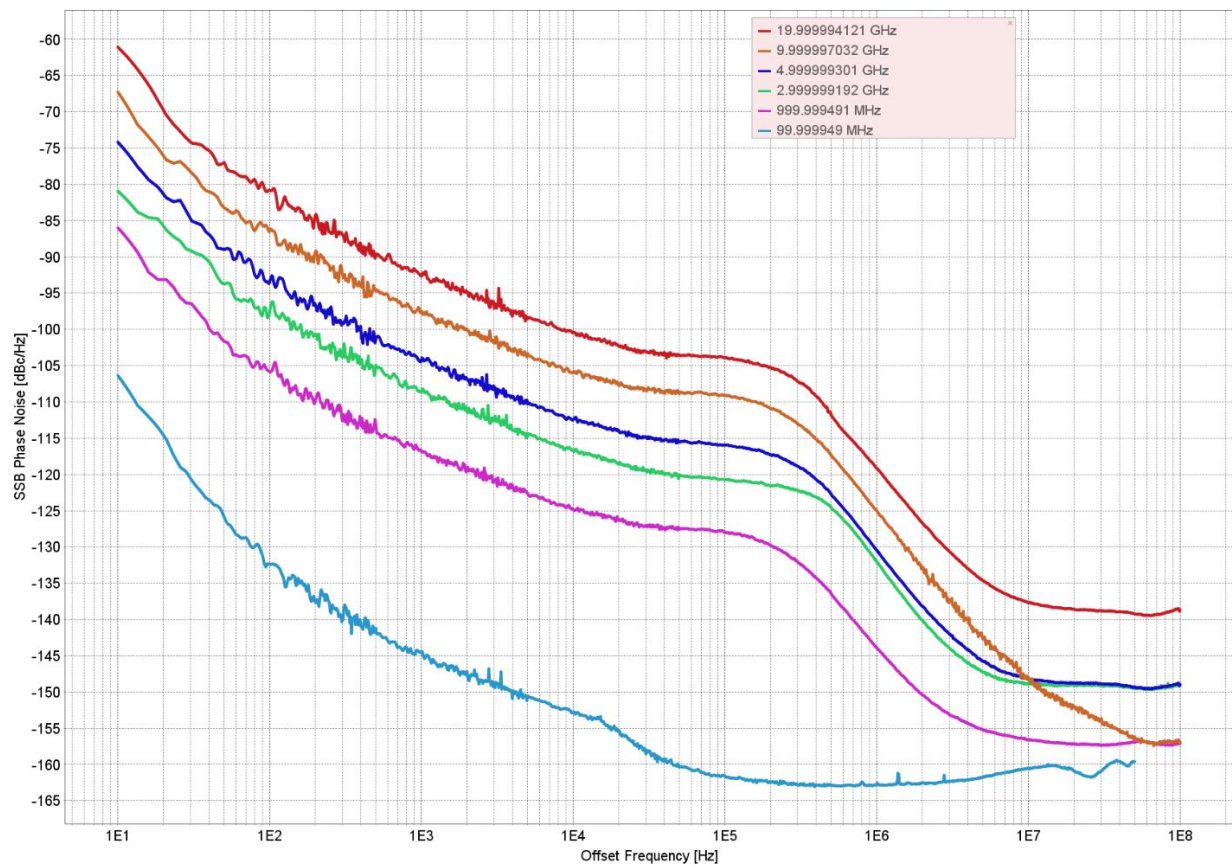
### Frequency Parameters / Range

| PARAMETER        | MIN              | TYPICAL                  | MAX                       | NOTE                       |
|------------------|------------------|--------------------------|---------------------------|----------------------------|
| Frequency Range  | 100 kHz<br>8 kHz |                          | 20 GHz<br>20 GHz          | Option 8K                  |
| Resolution       |                  | 0.01 Hz                  |                           |                            |
| Phase Resolution |                  | 0.1 deg                  |                           |                            |
| Switching Speed  |                  | 200 $\mu$ s<br>5 $\mu$ s | 500 $\mu$ s<br>10 $\mu$ s | In sweep mode<br>Option FS |

### Phase Noise

| PARAMETER                | MIN | TYPICAL     | MAX | NOTE           |
|--------------------------|-----|-------------|-----|----------------|
| SSB Phase noise at 1 GHz |     |             |     | See also plots |
| At 10 Hz from carrier    |     | -85 dBc/Hz  |     |                |
| At 1 kHz from carrier    |     | -115 dBc/Hz |     |                |
| At 20 kHz from carrier   |     | -125 dBc/Hz |     |                |
| At 10 MHz from carrier   |     | -155 dBc/Hz |     |                |

Figure 1: Phase Noise Performance



## Spectral Purity

| PARAMETER                    | MIN | TYPICAL | MAX     | NOTE                        |
|------------------------------|-----|---------|---------|-----------------------------|
| <b>Output Harmonics</b>      |     |         |         | $P_{out} = 0 \text{ dBm}$   |
| < 3.0 GHz                    |     | -15 dBc | 0 dBc   |                             |
| 3.0 – 7.5 GHz                |     | -25 dBc | -10 dBc |                             |
| 7.5 – 12.0 GHz               |     | -30 dBc | -20 dBc |                             |
| > 12.0 GHz                   |     | -40 dBc | -30 dBc |                             |
| <b>Sub-harmonics</b>         |     |         |         | $P_{out} = 0 \text{ dBm}$   |
| < 10.0 GHz                   |     | -80 dBc | -50 dBc |                             |
| 10.0 – 18.0 GHz              |     | -55 dBc | -40 dBc |                             |
| >18.0 GHz                    |     | -40 dBc | -20 dBc |                             |
| <b>Non-harmonic spurious</b> |     |         |         | (10 kHz < offset < 500 MHz) |
| <= 18.0 GHz                  |     | -65 dBc | -50 dBc |                             |
| > 18.0 GHz                   |     | -55 dBc | -35 dBc |                             |

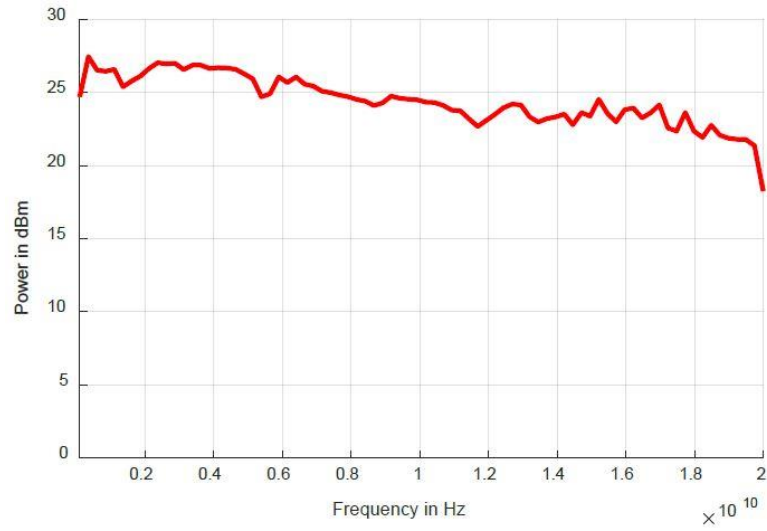
Figure 2: Harmonic performance at 0 dBm – Harmonic output power [dBc] vs. frequency [Hz]



## Level Performance

| PARAMETER  | MIN   | TYPICAL | MAX     | NOTE                       |
|------------|-------|---------|---------|----------------------------|
| Range      | 0 dBm |         | +18 dBm | Settable to -10 to +23 dBm |
| Resolution |       | 0.5 dB  |         |                            |

**Figure 3: Typical Maximum Output Power**



**Figure 4: Power Linearity**

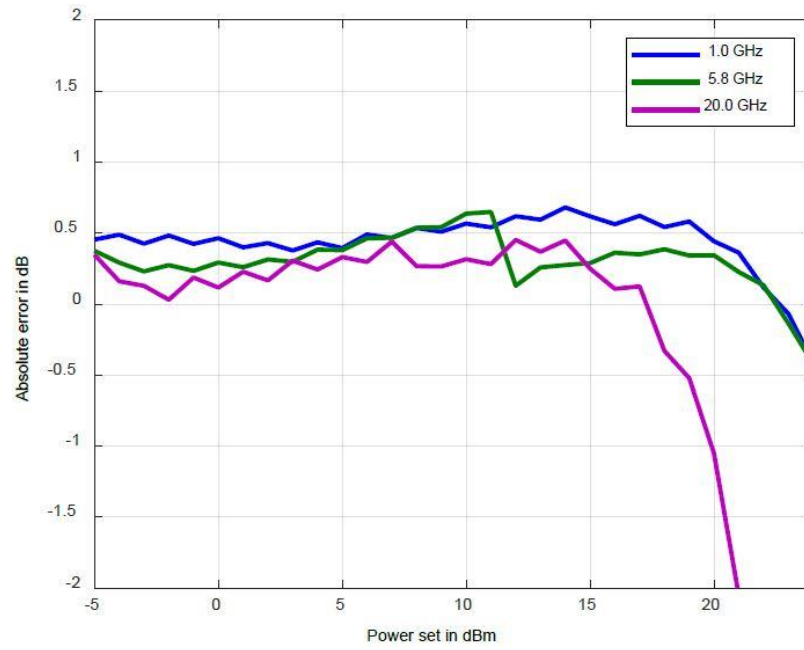
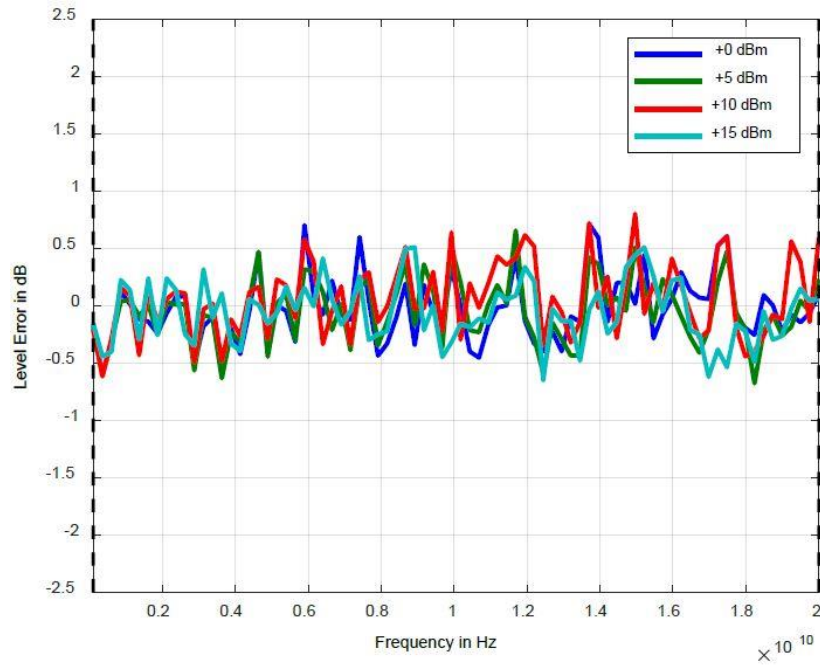




Figure 5: Power level accuracy



### Level Uncertainty and VSWR

| PARAMETER         | MIN | TYPICAL | MAX | NOTE |
|-------------------|-----|---------|-----|------|
| Level uncertainty |     | ±1.5 dB |     |      |
| Output impedance  |     | 50 Ω    | 2.0 |      |
| VSWR              |     | 1.7     |     |      |

### Reverse Power Protection

| PARAMETER  | MIN | TYPICAL | MAX    | NOTE |
|------------|-----|---------|--------|------|
| DC Voltage |     | 7 V     |        |      |
| RF Power   |     |         | 23 dBm |      |

### Channel to Channel Performance

| PARAMETER                | MIN   | TYPICAL | MAX | NOTE                |
|--------------------------|-------|---------|-----|---------------------|
| Isolation                |       |         |     |                     |
| < 3.0 GHz                | 90 dB |         |     |                     |
| 3.0 – 8.0 GHz            | 70 dB |         |     |                     |
| > 8.0 GHz                |       | 60 dB   |     |                     |
| Relative phase stability |       | 15 mrad |     | @5 GHz over 5 hours |

Figure 6: Channel to channel isolation – the measurement shows the impact of channel #2, #3 and #4 at  $f_0+9$  MHz on the channel #1 channel under test) operating at  $f_0$ . All channels have dBm output power.

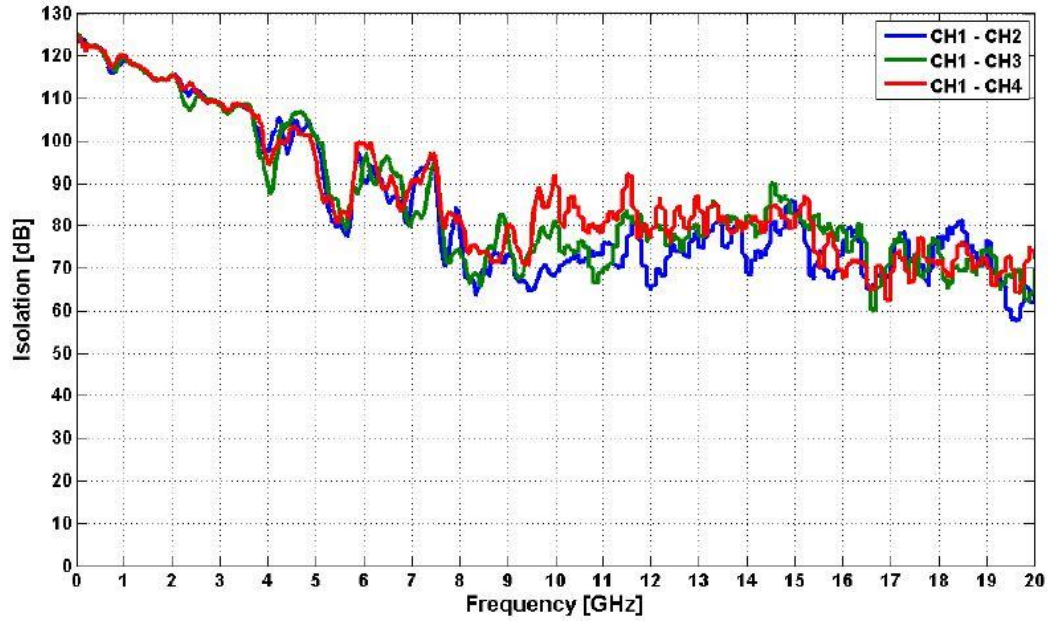


Figure 7: Channel to channel phase stability – the measurement shows the phase fluctuation between two RF channels in the same device, measured over 5 hours with a 5 GHz CW signal.

Model 825-M typical time domain channel-to-channel phase error at 5.000 GHz

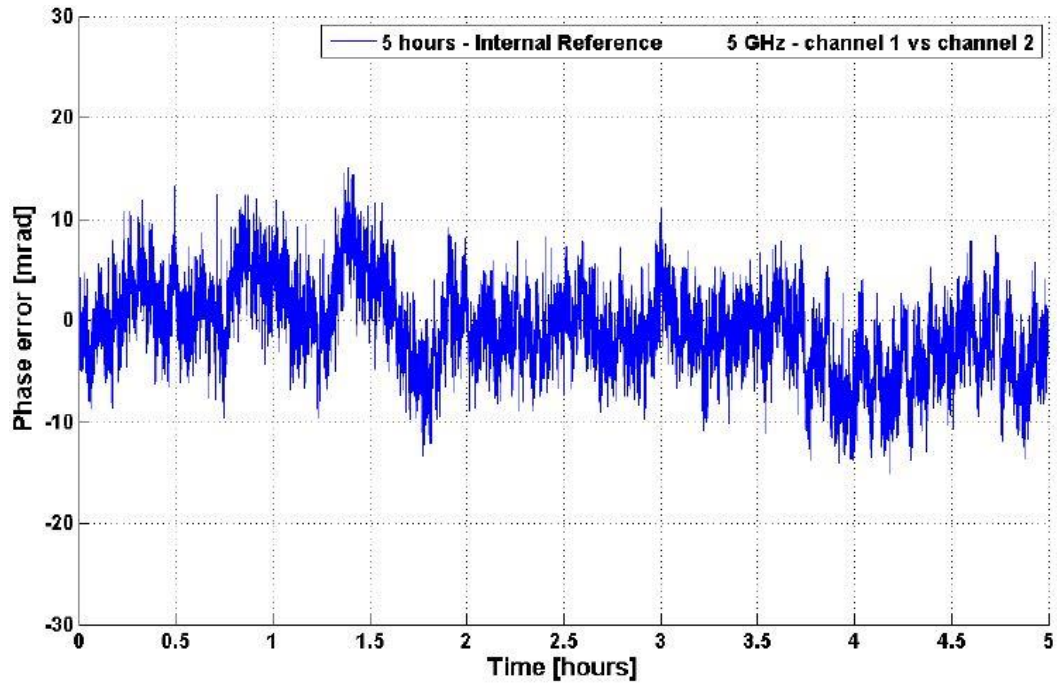




Figure 8: 825-M typical time domain channel-to-channel phase error at 10 GHz – measured over 12 hours

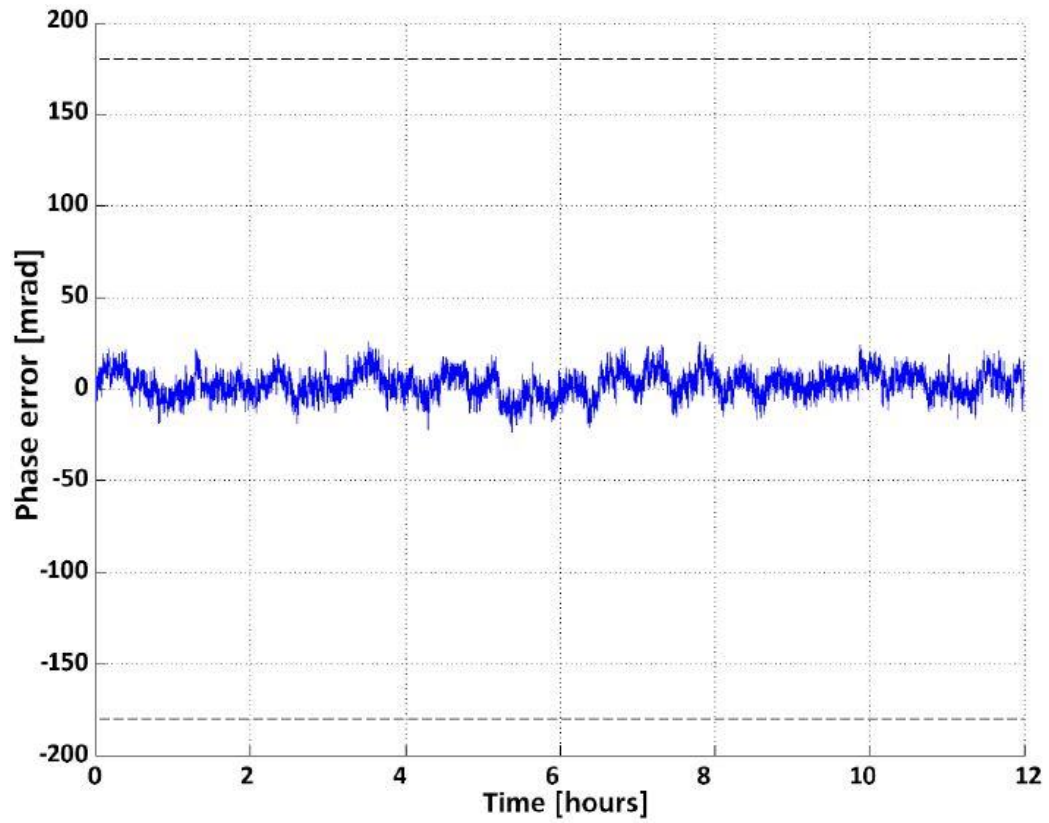
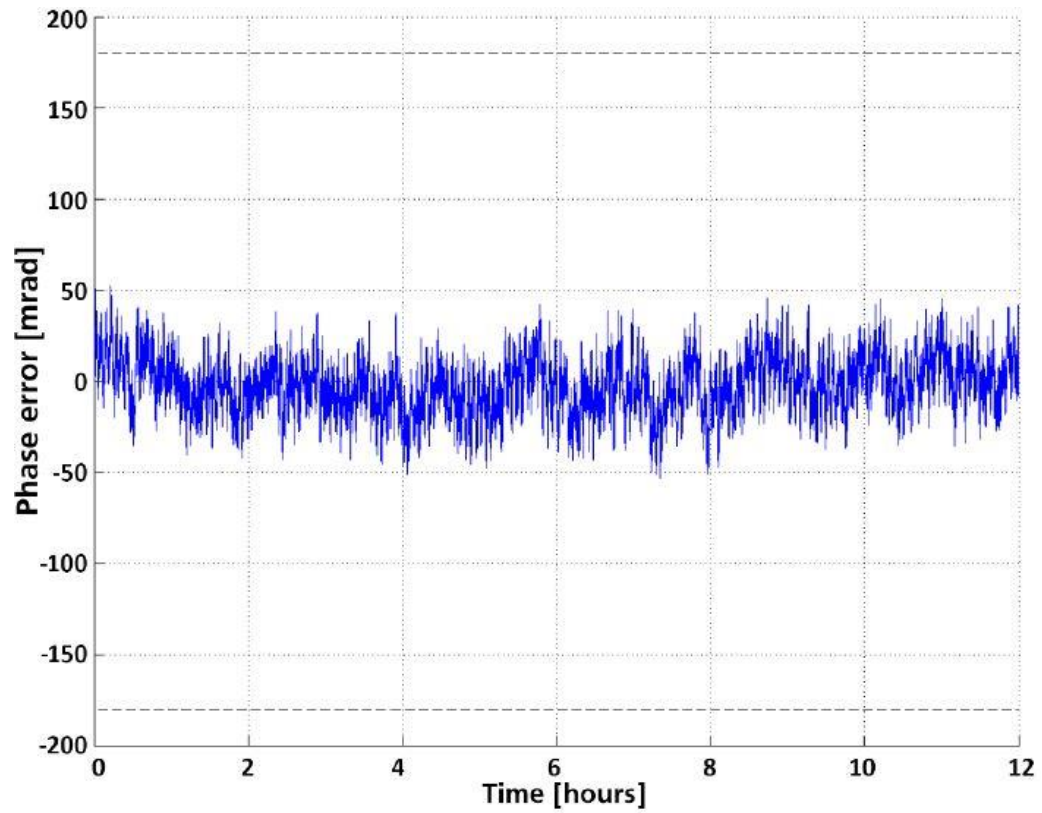


Figure 9: 825-M typical time domain channel-to-channel phase error at 20 GHz – measured over 12 hours



## Modulation Capabilities

| PARAMETER                    | MIN    | TYPICAL     | MAX    | NOTE           |
|------------------------------|--------|-------------|--------|----------------|
| <b>Pulse Modulation</b>      |        |             |        |                |
| On/off ratio                 |        | 60 dB       |        |                |
| Repetition frequency         | DC     |             | 10 MHz |                |
| Pulse width                  | 30 ns  |             | 20 s   |                |
| Pulse rise/fall time         |        | 9 ns        |        |                |
| Pulse trainlength (pulses)   | 2      |             | 4192   |                |
| Video crosstalk              |        | -40 dB      |        |                |
| Modulation source            |        | Int. / ext. |        |                |
| External input threshold     | 0.85 V | 0.9 V       | 0.95 V | TTL compatible |
| External input voltage range | -0.5 V |             | +5.5 V | TTL compatible |
| External input hysteresis    |        | 60 mV       |        |                |
| Delay (to RF)                |        | 20 ns       | 40 ns  |                |

## Sweeping Capability

| PARAMETER                                                                          | MIN                      | TYPICAL | MAX   | NOTE      |
|------------------------------------------------------------------------------------|--------------------------|---------|-------|-----------|
| <b>Frequency / List sweep</b>                                                      |                          |         |       |           |
| Sweep type: linear, logarithmic, random                                            |                          |         |       |           |
| Step time                                                                          | 500 $\mu$ s<br>5 $\mu$ s |         | 200 s | Option FS |
| Timing resolution                                                                  |                          | 5 ns    |       |           |
| Timing accuracy per point                                                          |                          | 20 ns   |       |           |
| <b>Generalized list sweep</b>                                                      |                          |         |       |           |
| Allows for individual setting of frequency, step-time, and off-time for each point |                          |         |       |           |

## Frequency Reference

| PARAMETER                                  | MIN    | TYPICAL               | MAX           | NOTE                        |
|--------------------------------------------|--------|-----------------------|---------------|-----------------------------|
| <b>Reference frequency input</b>           | 1 MHz  |                       | 200 MHz       | Integer MHz                 |
| Reference input level                      | -5 dBm | 0 dBm                 | +13 dBm       |                             |
| Lock Range                                 |        |                       | $\pm 1.0$ ppm |                             |
| Reference input impedance                  |        | 50 $\Omega$           |               |                             |
| <b>Internal reference frequency output</b> |        | 100 MHz               |               |                             |
| Initial accuracy of internal reference     |        | $\pm 40$ ppb          |               | calibrated at $23 \pm 3$ °C |
| Temperature stability (0 to 50 degC)       |        |                       | $\pm 100$ ppb |                             |
| <b>Aging</b>                               |        |                       |               |                             |
| Aging 1 <sup>st</sup> year                 |        | 0.5 ppm               |               |                             |
| Aging per day                              |        |                       | 5 ppb         | After 30 days operation     |
| Warm-Up time                               |        | 5 min                 |               |                             |
| Output of internal reference               |        | +0 dBm<br>50 $\Omega$ |               |                             |

| PARAMETER                            | MIN        | TYPICAL         | MAX    | NOTE                                                |
|--------------------------------------|------------|-----------------|--------|-----------------------------------------------------|
| Trigger Types                        |            |                 |        | Continuous, single (point), gated, gated direction  |
| Trigger Source                       |            |                 |        | External, bus (LAN, USB)                            |
| Trigger Modes                        |            |                 |        | Continuous free run, trigger and run, reset and run |
| Trigger uncertainty                  |            | 5 $\mu$ s       |        |                                                     |
| External Trigger delay               | 50 $\mu$ s |                 | 40 s   |                                                     |
| External Delay resolution            |            | 15 ns           |        |                                                     |
| Trigger Modulo                       | 1          |                 | 255    | Execute only on Nth trigger event                   |
| Trigger Polarity                     |            | Rising, falling |        |                                                     |
| External trigger input threshold     | 0.85 V     | 0.9 V           | 0.95 V | TTL compatible                                      |
| External trigger input voltage range | -0.5 V     |                 | +5.5 V | TTL compatible                                      |
| External trigger input hysteresis    |            | 60 mV           |        |                                                     |

\*TRIG OUT connector is not available in standard flange-mount enclosure

## CONNECTORS

### Front Panel and Rear Panel



#### Front View



### Rear View



### With Option HI



### Fast Control Port (FCP)

- 8-bit or 16-bit parallel port for fast, time critical settings like frequency
- Sequential submission of 48-bit frequency word or access to pre-defined frequency table
- Optional amplitude control and support for multi-channel models (only with 16-bit bus)
- Signal Source confirms the received data with ACK (only in 8-bit mode) and informs the controller by the BUSY signal while processing the information

**Connector:** 26 pin 3M Mini-D Ribbon Receptacle

**8-bit Mode:** Address A<3..0>, Data D<3..0>, STROBE, ACK, BUSY

**16-bit Mode:** Address A<7..0>, Data D<7..0>, STROBE, BUSY

**Input signal:** 0 to 5 V

**Input impedance:** 4,7 kΩ

**Maximum toggle rate:** 10 MHz, frequency switching starts after transfer of last byte

## ORDERING INFORMATION

| HOST MODEL | PRODUCT         | DESCRIPTION                                                   |
|------------|-----------------|---------------------------------------------------------------|
| 825-M      | 825-M           | 20 GHz wideband frequency synthesizer, flange-mount           |
| 825-M      | 825-M           | Multi-channel 20 GHz frequency synthesizer, 19" 1U rack-mount |
| 825-M      | Option EB       | External power bank adapter cable                             |
| 825-M      | Option 8K       | Frequency range extension to 8 kHz                            |
| 825-M      | Option FS       | Fast switching option (with FCP port)                         |
| 825-M      | Option HI       | High Isolation Enclosure                                      |
| 825-M      | Option GPIB     | GPIB interface (only with option 1U rack-mount)               |
| 825-M      | Option FLASH    | MicroSD card slot for removable SD memory                     |
| 825-M      | Option Retrofit | Applies when options are back-ordered                         |

## GENERAL CHARACTERISTICS

### Remote Programming Interfaces:

Ethernet interface  
 USB2.0 device interface  
 GPIB (option)  
 Control language: SCPI Version 1999.0

**Power requirements:** 24 VDC; 20 W maximum

**Mains adapter supplied:** 100-240 VAC in / 24 V, 2 A DC out

**Storage temperature range:** - 40 to 70 °C

**Operating temperature range:** 0 to 45 °C

**Operating and storage altitude:** Up to 15,000 feet



Safety/EMC complies with applicable Safety and EMC regulations and directives.

### Dimensions (W x L x H) / Weight [Including Connectors]

|                             |                                                                  |
|-----------------------------|------------------------------------------------------------------|
| Standard Flange-Mount       | 4.13 x 10.63 x 2.36 in [105 x 270 x 60 mm] / ~2 lbs (≤1.0 kg)    |
| Model 825-M in 1U Rackmount | 16.85 x 18.39 x 1.73 in [428 x 467 x 44 mm] / ~21 lbs (≤10.0 kg) |

**Recommended calibration cycle:** 24 months



## Document History

| Version | Date       | Author | Notes                                                   |
|---------|------------|--------|---------------------------------------------------------|
| V10     | 2019-05-20 | Jk     | First release                                           |
| V11     | 2019-09-20 | Jk     | Added options                                           |
| V12     | 2020-02-27 | Jk     | Single and multi-channel version                        |
| V121    | 2020-03-25 | Jk     | Plots added                                             |
| V122    | 2020-05-08 | Jk     | FCP specification added                                 |
| V123    | 2020-06-01 | Db     | FCP details added                                       |
| V124    | 2020-10-20 | Jk     | Refined harmonic specs                                  |
| V125    | 2020-12-07 | Yg     | Added multi-channel specs and pictures                  |
| V126    | 2021-02-25 | Db     | Pulse and trigger input electrical specifications       |
| V127    | 2021-04-13 | Ah     | Environmental specification updated                     |
| V128    | 2022-08-22 | Ee, Db | Corrected dimensions, corrected harmonic specifications |
| V129    | 2023-05-30 | Mh     | Corrected harmonics and non-harmonic values             |
| V130    | 2023-06-14 | Mh, ee | New layout structure                                    |
| V131    | 2023-11-29 | Ap     | Added phase stability plots for 10 GHz and 20 GHz       |
| V132    | 2024-08-15 | AT     | Removing the touch option                               |

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## NOTES